



ACTIVE AND PASSIVE DISTRACTION INTERVENTIONS IN A PEDIATRIC EMERGENCY DEPARTMENT TO REDUCE THE PAIN AND ANXIETY DURING VENOUS BLOOD SAMPLING: A RANDOMIZED CLINICAL TRIAL

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Contribution to Emergency Nursing Practice

- Studies have reported that distractions are effective in reducing procedural pain and anxiety in children during venous blood sampling.
- This study has used 2 new distractions: 1 active (rotatable wooden toy) and 1 passive (toy wristband). The study contributes to other study results showing that active distraction is effective in reducing procedural pain and anxiety in children during venous blood sampling.
- Pediatric emergency nurses should have knowledge of the efficacy and use of toy wristband and rotatable wooden toy practices, which are easy to use in reducing procedural pain, fear, and anxiety during venous blood sampling, and these methods could be used as routine nursing practices.

Abstract

Introduction: Distraction is a method that is easy to use in emergency departments and effective in relieving procedural pain and anxiety. This study aimed to determine the effect of 2

new distraction methods—1 active distraction (rotatable wooden toy) and 1 passive distraction (toy wristband)—on procedural pain, fear, and anxiety in children during venous blood sampling.

Methods: This study was a randomized controlled experimental study. The sample consisted of 216 children aged 6 years to 12 years. They were divided into 3 groups using the block randomization procedure: active distraction group ($n = 72$); passive distraction group ($n = 72$); and control group ($n = 72$). The levels of pain and anxiety in the children were measured before and during the blood sampling by the children themselves, their parents, and the researcher using the Visual Analog Scale, the Wong-Baker FACES Pain Rating Scale, and the Children's Fear Scale.

Results: The children and their parents included in the control and experimental groups had similar sociodemographic characteristics. The active distraction group had lower levels of procedural pain, fear, and anxiety than the other groups (children's visual analog scale score, $F = 134.22$; $P < 0.05$; Wong-Baker FACES Pain Rating Scale score, $F = 137.54$; $P < 0.001$; and Children's Fear Scale score, $F = 92.44$; $P < 0.001$).

Discussion: Both the toy wristband and rotatable wooden toy interventions can be used to reduce procedural pain, fear, and anxiety in children during blood sampling in emergency departments.

Key words: Anxiety; Active distraction; Blood sampling; Passive distraction; Procedural pain

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Introduction

Pain is a significant phenomenon that children frequently experience and is influenced by environmental, sociocultural, and individual factors, mostly causing fear, anxiety, and stress.^{1,2} One cause of this biopsychosocial emotion is an invasive medical procedure.^{3,4} Although venous blood sampling is thought to be a minor invasive procedure, it is

one of the greatest fears of children, and can turn into an unpleasant experience for children, parents, and health care professionals.^{1,5-7} Focusing on this issue, the American Academy of Pediatrics and the American Pain Society report that minor medical procedures such as blood sampling and vascular access should be performed properly, with a minimum level of pain and anxiety.⁸ Therefore procedural pain and anxiety assessment and management should be approached with careful attention in emergency departments.

There are many approaches, including pharmacological and nonpharmacological methods, to reduce procedural pain and anxiety that may occur during medical procedures.⁹⁻¹³ Nonpharmacological methods are independent nursing practices that are easy and cheap to use; are time-saving; have no adverse effects; promote collaboration in children; increase both activity level and sense of individual control; and reduce analgesic use, pain, stress, and anxiety.¹³ Distraction, a nonpharmacological method, is one of the easiest and most effective methods, which is used in 2 ways: active and passive distraction.^{9,11,12,14-16} Active distraction encourages children to participate in activities and display their own skills during medical procedures, activating their visual, auditory, and kinesthetic senses, whereas passive distraction is when parents or nurses involved in the medical procedures do not allow children to engage in activities, activating only their visual and auditory senses.^{9,11,12,14,15} A national survey across 15 pediatric emergency departments identified distraction as a procedure that could be easily implemented to manage children's pain.⁶ In addition, distractions used during medical procedures are effective in reducing procedural pain and anxiety in children in emergency departments.^{2,4,5,7} However, only one of the active and passive distractions is generally used in the same study and there are a limited number of studies showing the comparative effects of these 2 methods in controlling pain and anxiety. Therefore, there is a need for further research on this subject. This study examined the efficacy of 2 new distraction methods—1 active and 1 passive—in reducing procedural pain, fear, and anxiety in children aged 6 years to 12 years during routine venous blood sampling in the pediatric emergency department.

STUDY HYPOTHESES

- Hypothesis 1: The use of a rotatable wooden toy during venous blood sampling relieves procedural pain, fear, and anxiety in children.

- Hypothesis 2: The use of a toy wristband during venous blood sampling relieves procedural pain, fear, and anxiety in children.
- Hypothesis 3: The active distraction method is more effective than the passive distraction method.

Methods

This was a randomized controlled experimental study conducted to examine the effects of 2 new nonpharmacological distraction methods—1 active (rotatable wooden toy) and 1 passive (audible, colorful toy wristband) distraction—on procedural pain, fear, and anxiety in children during venous blood sampling.

STUDY POPULATION AND SAMPLES

This study was conducted between March 1, 2019, and April 30, 2019, in the blood sampling unit of the Pediatric Emergency Department at Dr. Sami Ulus Maternity, Children's Health, and Diseases Training and Research Hospital, which is one of the largest pediatric emergency departments in Ankara, Turkey. The study population consisted of children aged 6 years to 12 years who were admitted to the unit between these dates.

SAMPLE SIZE JUSTIFICATION

A power analysis was performed using G*Power version 3.1.9 (Heinrich Heine University, Dusseldorf, Germany). A study by Canbulat et al¹⁷ was taken as a reference. The sample size was determined as one that had a test power of 98% with an effect size of 30% at 95% confidence interval and with a Type 1 error probability (significance level) of 5% for the F-test analysis. Accordingly, the sample consisted of 216 children (active distraction group = 72, passive distraction group = 72, and control group = 72).

INCLUSION CRITERIA

The study inclusion criteria were as follows: children aged 6 years to 12 years; both children and parents were able to speak Turkish; having no mental or physical disability and no sedative, analgesic, or narcotic substance use 24 hours before admission to the blood sampling unit; blood sample being taken at the first attempt; and having a fever of less than 38°C (100.4°F). Fever can cause anxiety in children. They may not be as active or talkative as usual. In addition,

fever can lead to a temporary diminishment of mental abilities and changes in consciousness (fainting or swooning), and trigger convulsions. Because of these reasons, they were excluded from the study.

DATA COLLECTION TOOLS

Data were collected using an introductory information form, the visual analog scale (VAS), the Wong-Baker FACES Pain Rating Scale (WB), and the Children's Fear Scale (CFS).

INTRODUCTORY INFORMATION FORM

This form was prepared by the researchers in line with the literature.¹⁰ The form consisted of 15 questions regarding the introductory characteristics of the children and their parents (child's age, gender, parents' age, education levels, income levels, and so on).

VISUAL ANALOG SCALE

The visual analog scale is an easy-to-understand and applicable pain scale for children aged 3 years to 18 years.¹⁸ The scale is a 10-cm-long unidimensional ruler, ranging from "no pain" on the left-hand side to "unbearable pain" on the right-hand side. The child is asked to mark a point on the line that will identify their pain.¹⁹ The Turkish validity and reliability study was conducted by Aydın et al.¹⁹ The Cronbach's alpha coefficients of the scale ranged from 0.73 to 0.93.¹⁹

WONG-BAKER FACES PAIN RATING SCALE

This scale was developed by Wong and Baker²⁰ and is used to assess pain in children aged 3 years to 18 years. The scale consists of 6 faces representing the increasing severity of pain from left to right, and scored from 0 to 10 points (0 = no pain, 10 = very severe pain). The child is asked to choose the face expressing their pain.²⁰ The Cronbach's alpha coefficient of the scale was 0.93.²¹

CHILDREN'S FEAR SCALE

This scale was developed by McMurtry et al.²² to measure children's fear and anxiety levels. The CFS is suitable for children aged 3 years to 16 years. In this method, the child is shown a picture containing 5 facial expressions, which are scored from 0 (no fear and anxiety) to 4 (highest fear and anxiety). The scales' Turkish validity and reliability study

was conducted by Gerçekler et al.²³ The Cronbach's alpha coefficient of the scale was 0.89.²³

RESEARCH PROCEDURES

The CONSORT 2010 report²⁴ was used while planning and reporting the present study (Figure 1). Before the blood sampling procedure, all parents and children were informed about the study; written consent was obtained from the parents who wanted to participate in the study, whereas verbal and written assents were obtained from the children. Parents are the ones who know their children best, and they may notice sensitive changes in their child's attitudes and behaviors, and therefore they were included in the study. In this manner, we aimed to prevent bias and improve the validity of the data in the study. The introductory information was collected in approximately 3 minutes using the introductory information form. The parents and children were given explanations regarding the scales to be used in the study. The children were asked to score their pain levels on the VAS and WB and their fear and anxiety levels on the CFS. The parents and the researcher also scored the children's pain levels on the WB and fear and anxiety levels on the CFS by observing their behavioral status before the procedure.

ALLOCATION

In the study groups, a block (stratified) randomization method was used according to the child's age (from 6 years to 12 years), gender (female or male), and fear of blood sampling (afraid or not afraid). A total of 36 pouches were created (6 y, F, scared; 6 y, M, afraid; 6 y, F, not afraid; 6 y, M, not afraid; and so on), and there were 6 envelopes with different colors (2 for control, 2 for active distraction, and 2 for passive distraction) in each pouch. The children who agreed to participate in the study were assigned to the experimental or control group on picking the appropriate envelopes with different colors according to their characteristics. Each selected envelope was taken out of the pouch. The children were given an explanation regarding the blood sampling process, taking into consideration their developmental stage.

VENIPUNCTURE

The blood sampling was carried out by the researcher who had been working in the pediatric emergency department as a pediatric emergency nurse for 9 years. In this study, the participants underwent the blood sampling only when the research nurse was on duty. The children's pain, fear, and anxiety levels were measured during the blood sampling

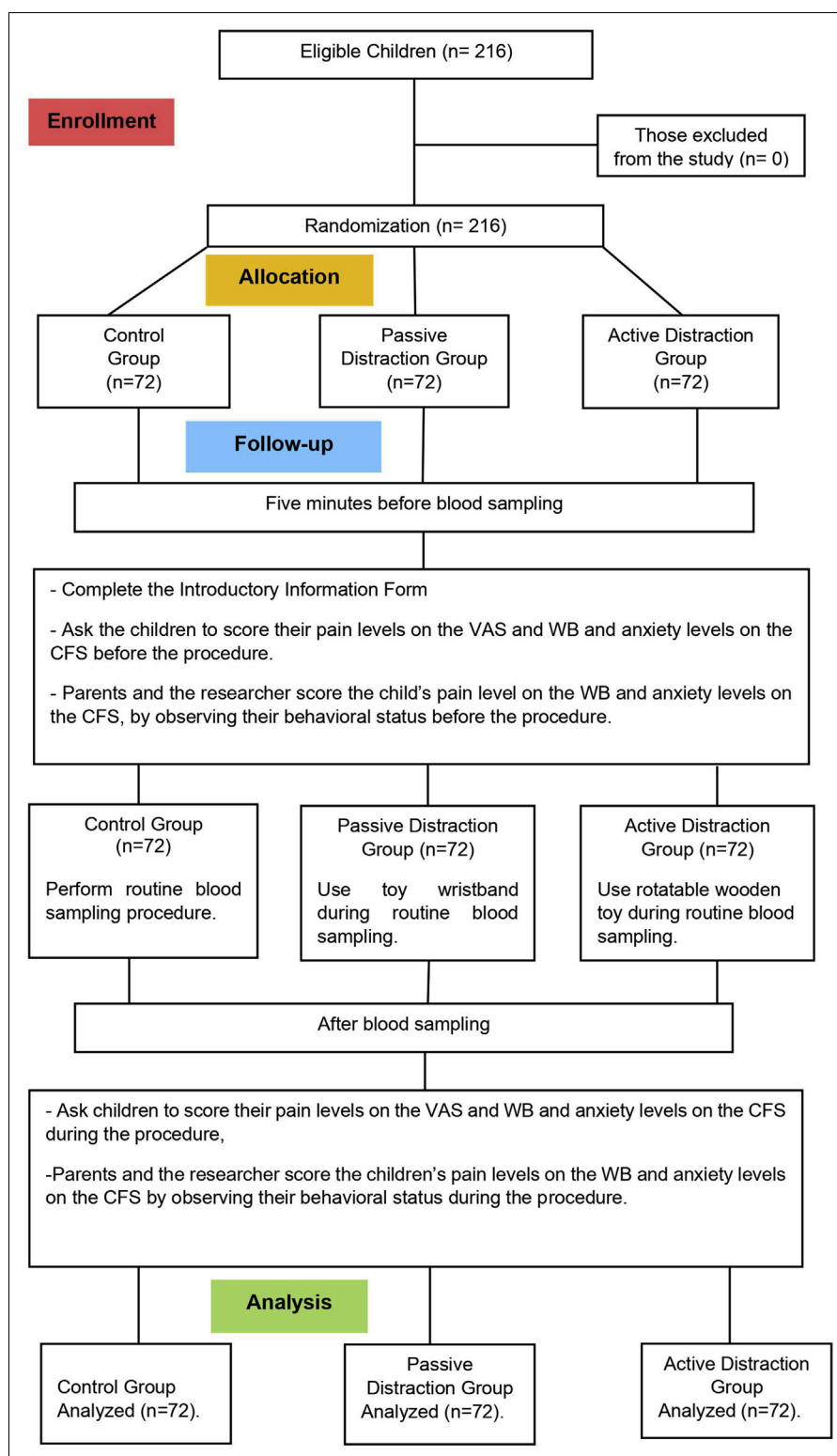


FIGURE 1

Study plan. VAS, visual analog scale; WB, Wong-Baker FACES Pain Rating Scale; CFS, Children's Fear Scale.

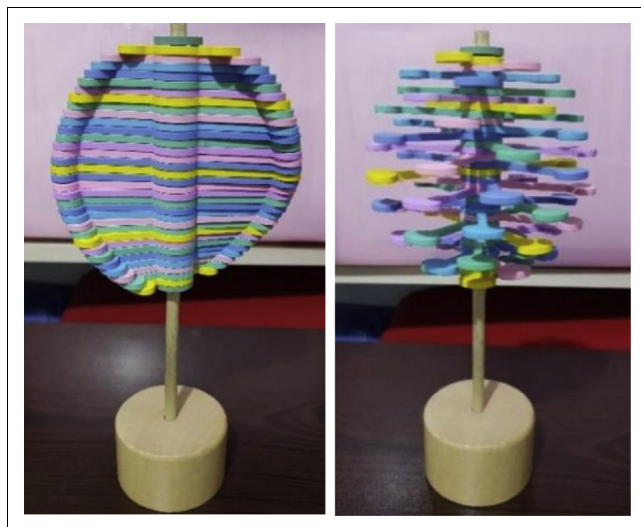


FIGURE 2
Rotatable wooden toy.

using the same procedure used for measuring their pain, fear, and anxiety levels before the blood sampling. The venous blood sampling was successfully completed in the first attempt for all groups.

CONTROL GROUP

The standard blood sampling procedure of the clinic was used for the children in the control group.

ACTIVE DISTRACTION GROUP

The rotatable wooden toy used for active distraction was produced by Edmark in 2008.²⁵ When its handle is spun rapidly, the toy transforms from a “DNA helix” into a “pine-cone” and back again in 1 motion owing to its kinetic shape

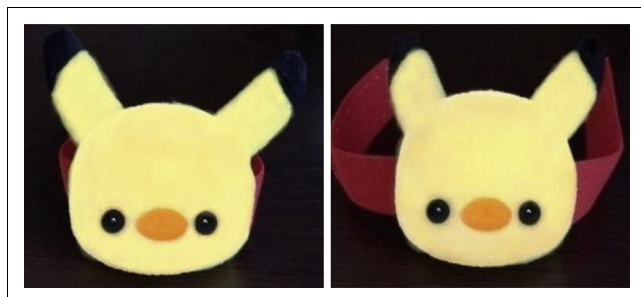


FIGURE 3
Audible stuffed toy wristband.

(Figure 2). The rotatable wooden toy is a toy that stimulates children’s cognitive, visual, and kinesthetic senses, enabling them to display their own skills. The children were instructed not to move the arm from which the blood was being drawn. They were asked to hold the toy with their free hand and spin its handle using their thumb and forefinger. They were encouraged to stay focused on spinning the toy until the sampling was over. The toy was disinfected after each procedure, making it ready for the next patient.

PASSIVE DISTRACTION GROUP

The toy wristband used for passive distraction is an audible, colored toy designed by the researcher. It consists of 2 parts: a 7-cm × 7-cm colored plush toy containing a sound device, and an elastic fabric wristband (Figure 3). When the top of the toy is pressed, it plays a melody twice. The wristband was attached loosely to the wrist of the child’s free arm so that the child could see the toy. The parent was asked to press the top of the toy to play a melody during the blood sampling procedure. The child was encouraged to look at the toy and listen to the melody until the blood sampling was over. The toy wristband was disinfected after each procedure, making it ready for the next patient.

ETHICAL CONSIDERATION

The Ethics Committee of Ankara University gave ethical approval (number 56786525-050.04.04/14249; February 25, 2019), and Dr. Sami Ulus Maternity, Children’s Health, and Diseases Training and Research Hospital gave institutional permission (number 73799008; March 12, 2019). Written consent was obtained from the parents, and verbal and written assents were obtained from the children.

DATA ANALYSIS

Data were evaluated using SPSS version 24.0 (IBM Corp, Armonk, NY) and analyzed using percentage distributions, mean, SD, and minimum-maximum values. A *t* test was used to compare the normally distributed data of the experimental and control groups, chi-square test and Fisher exact test were used to examine the demographic differences in the experimental and control groups, analysis of variance was used to examine the differences between pain and anxiety mean scores of the control and experimental groups, and post hoc (Bonferroni) advanced analyses were used to make intragroup binary comparisons. *P* values less than 0.05 were regarded as statistically significant.

TABLE 1

Introductory characteristics of children and parents (N = 216)

Variables	Control group (n = 72) n (%)		Passive distraction group (n = 72) n (%)		Active distraction group (n = 72) n (%)		Test statistics χ^2	P value
Child's age								
6–9	36	(50)	36	(50)	36	(50)	<.001	0.99
10–12	36	(50)	36	(50)	36	(50)		
Child's gender								
Female	36	(50)	36	(50)	36	(50)	<.001	0.99
Male	36	(50)	36	(50)	36	(50)		
The parent accompanying the child								
Mother	41	(56.9)	44	(61.1)	46	(63.8)	0.74	0.69
Father	31	(43.1)	28	(38.9)	26	(36.2)		
Parent's age								
20–40	62	(86.1)	57	(79.1)	54	(75)	7.22	0.30
41 and above	10	(13.8)	15	(20.8)	18	(25)		
Parent's education								
Less than primary school	1	(1.38)	3	(4.1)	3	(4.1)	6.93*	0.54
Primary school	32	(44.4)	20	(27.7)	29	(40.2)		
High school	23	(31.9)	27	(37.5)	20	(27.7)		
University and above	16	(22.2)	22	(30.5)	20	(27.7)		

* Fisher exact test.

Results

RESULTS REGARDING THE INTRODUCTORY CHARACTERISTICS OF CHILDREN AND PARENTS

There were no significant differences in the children's and parent's demographic characteristics in both control and experimental groups ($P > 0.05$) (Table 1).

COMPARISON OF THE CHILDREN'S BLOOD SAMPLING EXPERIENCES

The number of children with and without an experience of blood sampling ($n = 71$ and $n = 1$, respectively) was randomly equal in the control and experimental groups ($P > 0.05$). There was no significant difference among their experiences of blood sampling (by number of blood sampling procedures, time of last blood sampling, and previous hospitalization) ($P > 0.05$). No nonpharmacological methods were used to reduce pain and anxiety during previous blood sampling procedures for any children in the control and experimental groups (Table 2).

COMPARISON OF THE WB MEAN SCORES MEASURED BEFORE AND DURING BLOOD SAMPLING

There was a significant difference in the mean WB pain scores between the control and active distraction groups before the blood sampling (children's WB score, $F = 3.42$; $P < 0.05$). However, there was a statistically significant difference among the WB mean scores of all groups measured during the blood sampling (children's WB score, $F = 137.54$; $P < 0.001$) (Table 3). The active distraction group had the lowest pain level, followed by the passive distraction and control groups, respectively (children's WB score, $F = 137.54$; $P < 0.001$). In addition, there was no statistically significant difference between the WB mean scores of the control and experimental groups measured by the 3 raters ($F = 1.007$; $P > 0.05$).

COMPARISON OF THE VAS PAIN MEAN SCORES MEASURED BEFORE AND DURING BLOOD SAMPLING

The control group had different VAS pain mean score measured before the blood sampling than the active distraction group ($F = 3.54$; $P < 0.05$). There was a statistically

TABLE 2

Comparison of the children's blood sampling experiences (N = 216)

Blood sampling experiences	Control group (n = 72) n (%)		Passive distraction group (n = 72) n (%)		Active distraction group (n = 72) n (%)		Test statistics χ^2	P value
Previous blood sampling procedures								
Yes	71	(98.6)	71	(98.6)	71	(98.6)	0.43*	0.99*
No	1	(1.4)	1	(1.4)	1	(1.4)		
Number of previous blood sampling procedures								
1–5	34	(47.2)	30	(41.6)	31	(43.1)	2.27	0.89
6–10	20	(28.6)	20	(28.6)	18	(25.7)		
11 or more	15	(21.4)	20	(28.6)	20	(28.6)		
Time of blood sampling procedures								
In last 6 mo	39	(54.1)	45	(62.5)	50	(69.4)	8.04	0.42
In last 1 y	16	(22.9)	7	(10.0)	8	(11.4)		
More than 1 y ago	14	(20.0)	17	(24.3)	11	(15.7)		
Fear of blood sampling								
Yes	36	(50)	36	(50)	36	(50)	<.001	0.99
No	36	(50)	36	(50)	36	(50)		
Previous hospitalization								
Yes	32	(45.7)	27	(38.6)	28	(38.6)	0.81	0.60
No	40	(54.3)	45	(61.4)	44	(61.4)		
Status of having a nonpharmacological intervention during blood sampling [†]								
Yes	—	—	—	—	—	—	—	—
No	72	(100)	72	(100)	72	(100)		

* Fisher exact test.

[†] Not included in the analysis.

significant difference in the VAS pain mean scores of all groups measured during the blood sampling. The active distraction group had the lowest pain level, followed by the passive distraction and control groups, respectively ($F = 134.22$, $P < 0.05$) (Table 4).

COMPARISON OF THE CFS ANXIETY MEAN SCORES MEASURED BEFORE AND DURING BLOOD SAMPLING

The levels of anxiety experienced by children before and during the blood sampling were evaluated using the CFS by the children themselves, their parents, and the researcher. There was no statistically significant difference in the CFS mean scores of all groups before blood sampling ($P > 0.05$). There was a statistically significant difference between the CFS mean scores of the control and experimental groups measured during the blood sampling (children's CFS score, $F = 92.44$; $P < 0.05$). The active distraction group

had the lowest anxiety level, followed by the passive distraction and control groups, respectively (Table 5). In addition, there was no statistically significant difference between the CFS mean scores of the control and experimental groups measured by the 3 raters ($F = 0.54$; $P > 0.05$).

Discussion

This study examined the efficacy of using the rotatable wooden toy and toy wristband to reduce procedural pain, fear, and anxiety in children during venous blood sampling in a pediatric emergency department by comparing active and passive distraction effects. Our results indicated that the toy wristband and rotatable wooden toy decreased procedural pain and anxiety, and that the active distraction method was more effective than the passive distraction method, supporting our 3 hypotheses. There is no study

TABLE 3

Comparison of the WB pain mean scores measured before and during blood sampling

Evaluation	Pain measurement time	Control group (n = 72) mean (SD)	Passive distraction group (n = 72) mean (SD)	Active distraction group (n = 72) mean (SD)	F	P value	Paired comparisons*
Child	Before blood sampling	5.49 (2.16)	5.86 (1.84)	6.34 (1.80)	3.424	<0.05	AD > CG; $P = 0.02$
	During blood sampling	7.33 (2.41)	3.30 (1.95)	2.60 (1.54)	137.538	<0.001	CG > PD > AD; $P < 0.001$
Parent	During blood sampling	5.51 (2.08)	6.00 (1.73)	6.46 (1.63)	4.649	<0.05	AD > CG*; $P < 0.05$
	During blood sampling	7.99 (1.93)	2.81 (1.72)	2.04 (1.20)	261.098	<0.001	CG > PD > AD; $P < 0.001$
Researcher	During blood sampling	5.46 (2.15)	6.06 (1.66)	6.46 (1.63)	5.270	<0.05	AD > CG; $P < 0.05$
	During blood sampling	7.94 (1.84)	2.79 (1.77)	1.79 (0.97)	307.525	< 0.001	CG > PD > AD; $P < 0.001$

CG, control group; PD, passive distraction group; AD, active distraction group.

* The groups causing significant difference according to Bonferroni post hoc advanced analysis.

TABLE 4

Comparison of the VAS pain mean scores measured before and during blood sampling

VAS rating time	Control group (n = 72) mean (SD)	Passive distraction group (n = 72) mean (SD)	Active distraction group (n = 72) mean (SD)	F	P value	Paired comparisons*
Before blood sampling	2.96 (0.97)	3.17 (0.76)	3.33 (0.73)	3.541	0.03	AD > CG; $P = 0.02$
During blood sampling	3.79 (1.08)	1.97 (0.81)	1.50 (0.65)	134.220	< 0.001	CG > PD > AD; $P < 0.001$

CG, control group; PD, passive distraction group; AD, active distraction group.

* The groups causing significant difference according to Bonferroni post hoc advanced analysis.

TABLE 5
Comparison of the CFS anxiety mean scores measured before and during blood sampling

Evaluation	Anxiety measurement time	Control group (n = 72) mean (SD)	Passive distraction group (n = 72) mean (SD)	Active distraction group (n = 72) mean (SD)	F	P value	Paired comparisons*
Child	Before blood sampling	3.21 (1.22)	3.36 (1.00)	3.60 (0.85)	2.45	0.08	—
	During blood sampling	3.91 (1.22)	2.09 (1.07)	1.63 (0.82)	92.44	< 0.001	CG > PD > AD; $P < 0.001$
Parent	Before blood sampling	3.29 (1.05)	3.50 (1.04)	3.56 (0.73)	1.21	0.20	—
	During blood sampling	4.19 (0.99)	2.09 (0.95)	1.61 (0.66)	166.95	< 0.001	CG > PD > AD; $P < 0.001$
Researcher	Before blood sampling	3.29 (1.05)	3.44 (0.97)	3.51 (0.81)	1.06	0.34	—
	During blood sampling	4.29 (0.90)	2.10 (0.95)	1.51 (0.60)	214.58	< 0.001	CG > PD > AD; $P < 0.001$

CG, control group; PD, passive distraction group; AD, active distraction group.

* The groups causing significant difference according to Bonferroni post hoc advanced analysis.

in which a toy wristband, as a passive distraction method, and a rotatable wooden toy, as an active distraction method, have been used during venous blood sampling or in any medical procedure. In this aspect, this is the first study specific to this subject in the field. Children in the control and experimental groups were similar in terms of age, gender, parents' ages, and education levels (Table 1). The similarity of their introductory characteristics indicated that blocking and randomization were successful. A previous blood sampling experience has the potential to affect children's pain perceptions and anxiety levels; therefore, the similarity of the groups in terms of these variables (Table 2) was important for the efficacy of the active and passive distraction methods used during the venous blood sampling.

Venous blood sampling is widely used in the diagnosis and treatment of various diseases,²⁶ but it may turn into a painful and stressful experience for children when adequate pain control cannot be achieved.²⁷ Therefore, procedural pain management is very important for pediatric emergency nurses working with children. This study found that both methods reduced pain (Tables 3 and 4), although active distraction was more effective in reducing pain. There are few studies comparing active and passive distractions in pain relief during blood sampling in emergency departments as well as other units, and they showed contradictory results. In addition to studies reporting that active distraction is more effective in pain relief than passive distraction,^{7,9,12,26} there are studies suggesting that passive distraction is more effective in pain relief than active distraction.^{10,11} Studies have reported that toys actively played with are effective in distracting children and reducing their pain levels by activating their visual, auditory, and kinesthetic senses during painful medical procedures.^{9,26,27} In this regard, the fact that the rotatable wooden toy was more effective in reducing procedural pain may be because this toy changes shape after a complete turn once it is given a simple and rapid spin. The changing shapes attract children's attention, and they need to focus on the toy to actively spin the handle. This is active distraction, which stimulates their cognitive, visual, and kinesthetic senses, which in turn reduces their procedural pain levels.

A fear of medical procedures may develop if adequate nursing care is not provided during invasive procedures.^{15,28} This can reduce children's participation in health care procedures, prevent them from receiving health services in case of illness, and negatively affect the treatment process by increasing their aggressive behaviors.¹⁵ Therefore, it is important to eliminate fear and anxiety during invasive procedures in emergency departments. This study found that both of these methods reduced fear and anxiety in the children during the blood sampling (Table 5), with active

distraction being more effective than passive distraction. There are very few studies comparing active and passive distractions in reducing fear and anxiety during blood sampling. Some of them found that active distraction was more effective in pain relief than passive distraction,^{9,10} whereas some others found that they had similar effects.^{2,7,11,26} There are also studies examining only one of these methods.^{29,30} This study found that the rotatable wooden toy was more effective in reducing the pain and anxiety of the children during the blood sampling. This may be because children are more interested in creating new and simple shapes,²⁷ so they direct their attention to a pleasant and exciting process and away from anxiety-inducing stimuli through a game-playing experience.^{9,26,29}

Limitations

There are several limitations to this study. First, the researcher was the person who collected both information and blood samples from the children. Because the researcher was a nurse working in the unit where the study was conducted, and because of the workload and inadequate number of nurses in the unit, the study could not be blinded. To prevent bias and increase the validity of the data, the children's pain and anxiety levels before and during the procedure were scored by the children themselves, their parents, and the researcher. They were blinded to each other's scores. Second, the reasons for the children's admission to the hospital were not examined, but the pain level before the procedure was evaluated as the factor affecting the study. Third, taking into account the possibility of contamination of the distracting devices, the wooden toy and toy wristband were disinfected with an alcohol-based disinfectant after each blood draw. The study protocol was not preregistered.

Implications for Emergency Nurses

Distraction is a nonpharmacological nursing intervention used for procedural pain and anxiety relief during blood sampling. According to the literature, the nonpharmacological methods used during blood collection in emergency departments include cartoon-patterned clothes,² bubble-blowing,² distraction kits,⁴ parental involvement and parental presence,⁵ Buzzy,⁷ distracting cards, and balloon-inflating.⁷ In Turkey, there is frequent crowding in emergency departments, and there is limited time for nonpharmacological methods and play. Most of the time, procedural pain management cannot be maintained effectively. As a matter of fact, in

our study the participants stated that nonpharmacological methods were not used to reduce pain and anxiety in previous medical procedures. They showed great interest in the study and interventions. Pediatric emergency nurses should be aware of the pain and anxiety of children in medical procedures and should attempt to reduce the pain and anxiety, although they have limited time at their disposal. During this process, distracting methods should be chosen according to the developmental characteristics of the child. In our study, the age groups were determined as ages 6 years to 9 years and ages 10 years to 12 years according to the gross and fine motor activities, cognitive development, and attention span of the children.³¹ The toy wristband and rotatable wooden toy were found to be effective in reducing pain and anxiety in both age groups. Hospitals should develop quality assurance procedures to improve the management of procedural pain and anxiety in emergency departments. Pediatric nurses should be informed with in-service trainings on the use and effectiveness of the toy wristband and rotatable wooden toy applications, which are easy to use and cost-efficient, and these methods should be used as routine nursing practice. Once these methods become well-known, future research can focus on implementing strategies to improve procedural pain and anxiety management in the pediatric ED setting.

Conclusions

Active and passive distraction techniques are effective in reducing pain, fear, and anxiety; therefore, they are recommended to be used during venous blood sampling. On the basis of the findings from this study, the toy wristband and rotatable wooden toy are feasible distraction interventions to potentially help manage procedural pain and anxiety in emergency departments. However, further evidence-based studies should be conducted to test the efficacy of these 2 new applications in painful procedures other than venous blood sampling and in children from different age groups.

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Author Disclosures

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